

Diffusion Models for Combinatorial Optimization

Undergraduate Thesis

Submitted in partial fulfillment of the requirements for the degree of
Bachelor of Science in Industrial Engineering

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2026

Abstract

Keywords: combinatorial optimization, traveling salesman problem, capacitated vehicle routing problem, diffusion models, reinforcement learning, neural combinatorial optimization.

Acknowledgments

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Chapter 1

Introduction

1.1 Motivation

Combinatorial optimization (CO) is a fundamental field in computer science and operations research, encompassing the search for an optimal solution from a finite but often exponentially large set of candidates. Classic examples include the traveling salesman problem (TSP), the maximum independent set (MIS), and the vehicle routing problem (VRP). These problems are NP-hard, meaning they are believed to be intractable in polynomial time, and no known algorithm can solve them efficiently at scale.

Traditionally, CO problems have been handled by exact solvers (e.g., Concorde and Gurobi) [1, 4], approximation algorithms, and hand-crafted heuristics such as LKH and local search methods like 2-opt [3, 5, 6]. However, these methods have significant drawbacks, often demanding substantial domain expertise, careful engineering, or considerable computational resources. These limitations have motivated the search for data-driven alternatives. In recent years, neural combinatorial optimization (NCO) has emerged as a promising approach to address the constraints of traditional methods [2].

As the field of NCO has progressed, various architectures have been proposed. In this paper, we utilize a diffusion-based architecture modeled with a Bernoulli distribution to solve CO problems for two primary reasons: 1) Diffusion-based architectures can capture high-dimensional, multi-modal distributions, evaluating solution probabilities simultaneously. Due to the inherent complexity of CO problems, this multi-modality is crucial for capturing the underlying problem structure. 2) The diffusion process iteratively refines its output, a mechanism that conceptually mirrors the iterative improvement approaches of classical solvers. Consequently, this research investigates methods to enhance the capabilities of diffusion-based models in solving CO problems.

1.2 Problem Statement

1.3 Contributions

- Reproduce and extend *DifusCO*-style diffusion solvers for TSP and CVRP.
- Implement *CADO*-style hybrid fine-tuning and REINFORCE on top of supervised checkpoints.
- Empirical comparison: optimality gap, wall-clock cost, and ablations (reward mode, entropy, denoising steps).

1.4 Thesis Outline

Chapter 2

Background

2.1 Combinatorial Optimization

2.2 Classical and Learning-Based Solvers

2.3 Diffusion Models for COP

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Chapter 3

Methodology

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Chapter 4

Experiments

- 4.1 Experimental Setup
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Chapter 5

Conclusion

5.1 Summary

5.2 Limitations

5.3 Future Work

Appendix A

Implementation Details

A.1 Configuration Reference

A.2 Hyperparameter Tables

A.3 Additional Plots

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